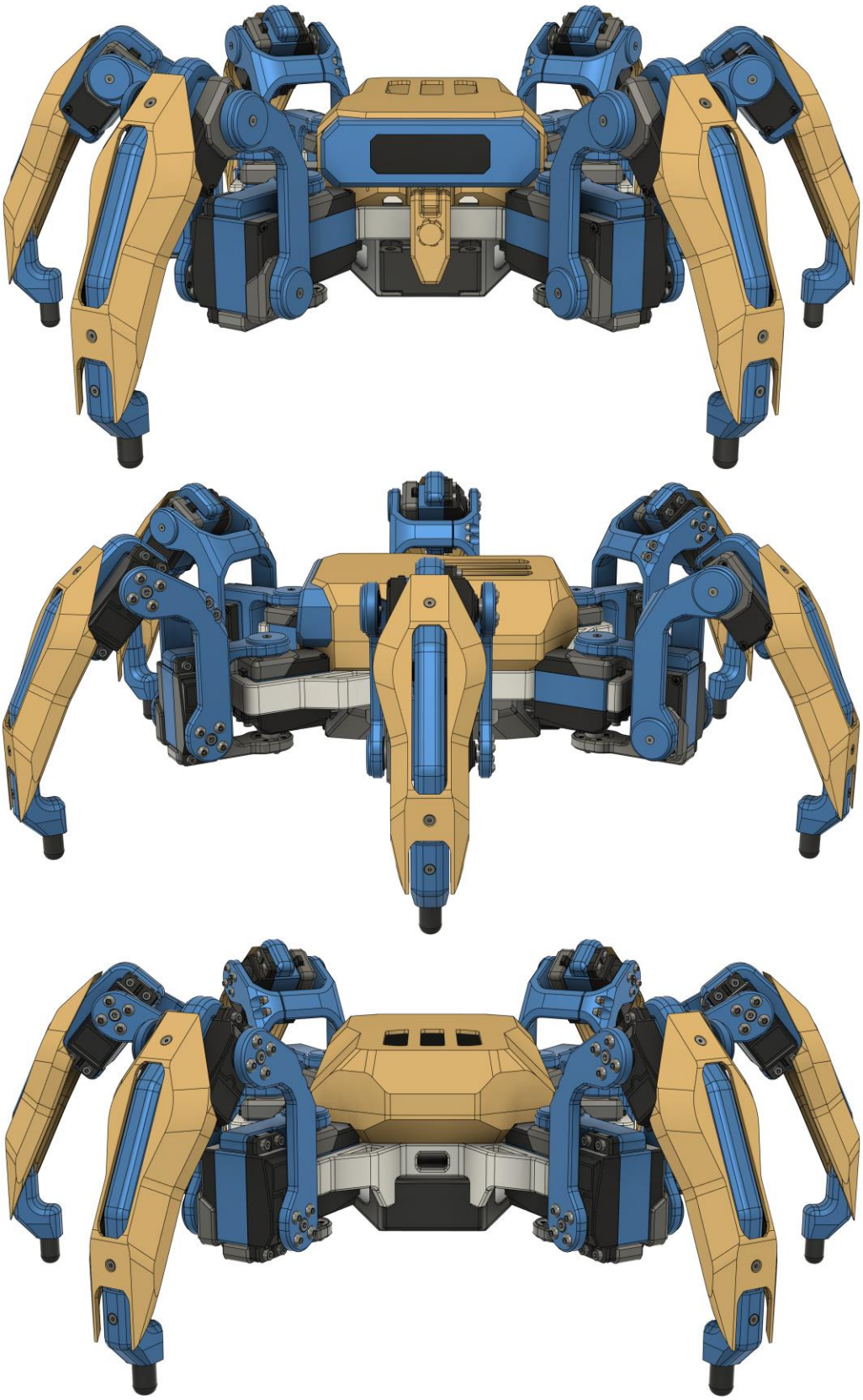
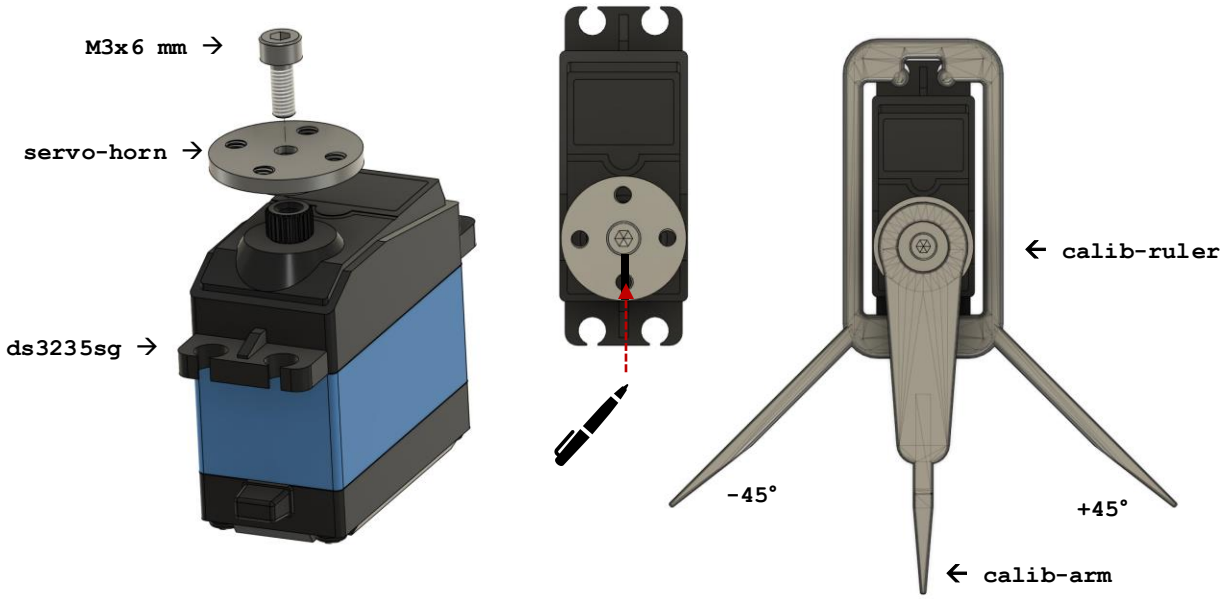


Notes

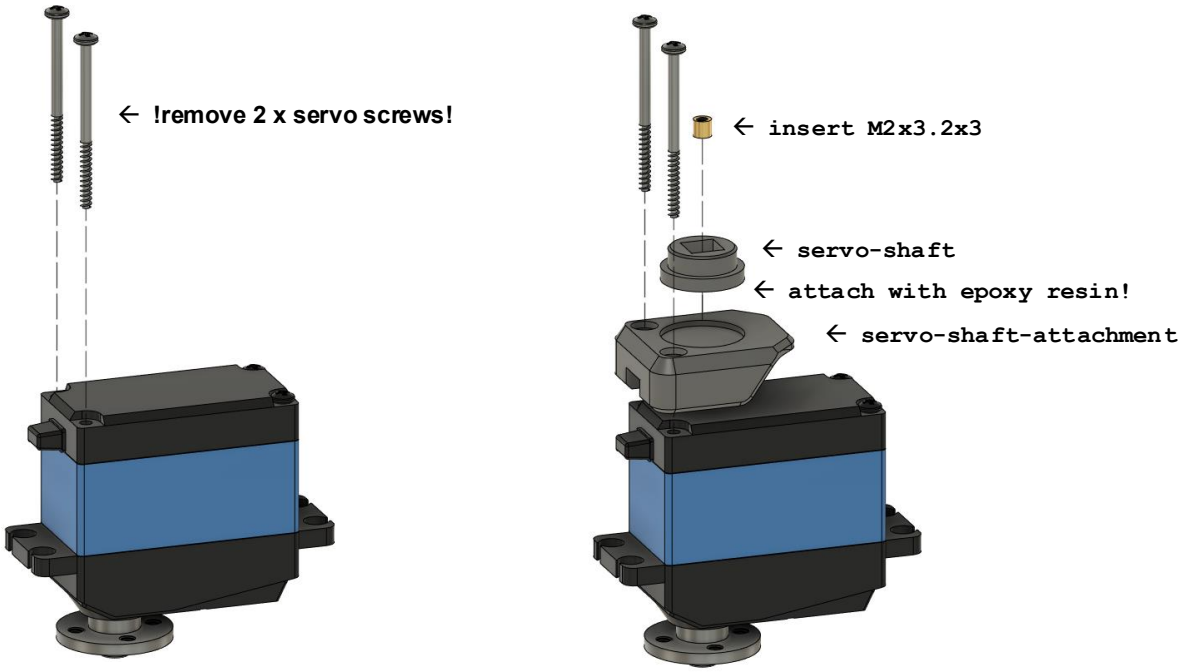
1) **This is a fairly complex open source DIY hexapod. The models, code and instruction are free. No support or quarantee is provided. Mistakes or errors in the BOM or instructions can be submitted via github issues.**
2) **To build the hexapod, you need the materials listed in the BOM and the following tools: M2, M2.5, M3 hex keys, 3D-printer and soldering iron. For servo calibration, either a manual servo calibrator or a pimoroni servo2040 board are required.**
3) **The instructions contain two alternative leg tip assemblies (D-1 and D-2). It is recommended to build the hexapod using the simpler leg tips. The sensors are currently _not_ used in the ROS 2 controller software. Only build the sensor legs if you plan to implement them in you own hexapod or you are coding your own locomotion/gait platform. The D-2 leg tip also SUCKS and needs a redesign!**
4) **Good luck and have fun!**

A. Servo calibration (x 18)

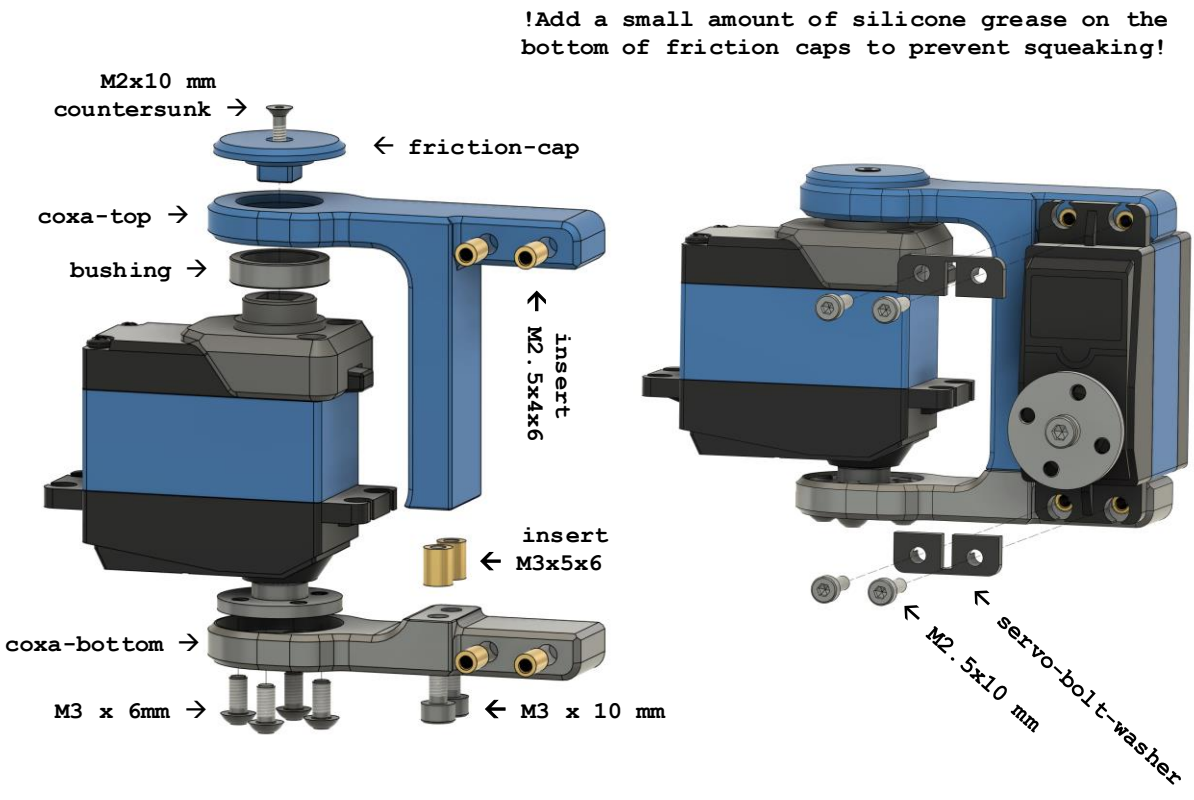
1) **Attach servo to servo controller or servo2040 (servoCalibration.uf2)**
2) **Move servo to the middle position (1500 ms).**
3) **Attach servo horn so that holes are aligned as pictured and mark the direction of the servo horn. Note: due to the way servos are assembled, you can typically never align the servo horn exactly, this is why we are doing the calibration.**
4) **Using either a separate servo calibrator or a servo2040 board with the servo calibration firmware loaded (recommended option), collect +45° and -45° values using a servo calibrator or a servo2040 with calibrations firmware. Make sure to mark the servos with numbers so that you can later match the calibrated values.**
5) **How to calibrate servos using the servo2040: https://github.com/olli-io/hexapod-servo2040-driver/blob/master/src/servoCalibration/README.md**



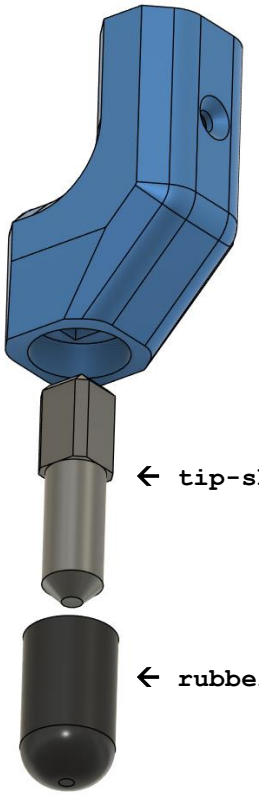
B. Servo assembly × 18



C. Coxa assembly × 6



D-1. Tip assembly × 6 (RECOMMENDED)

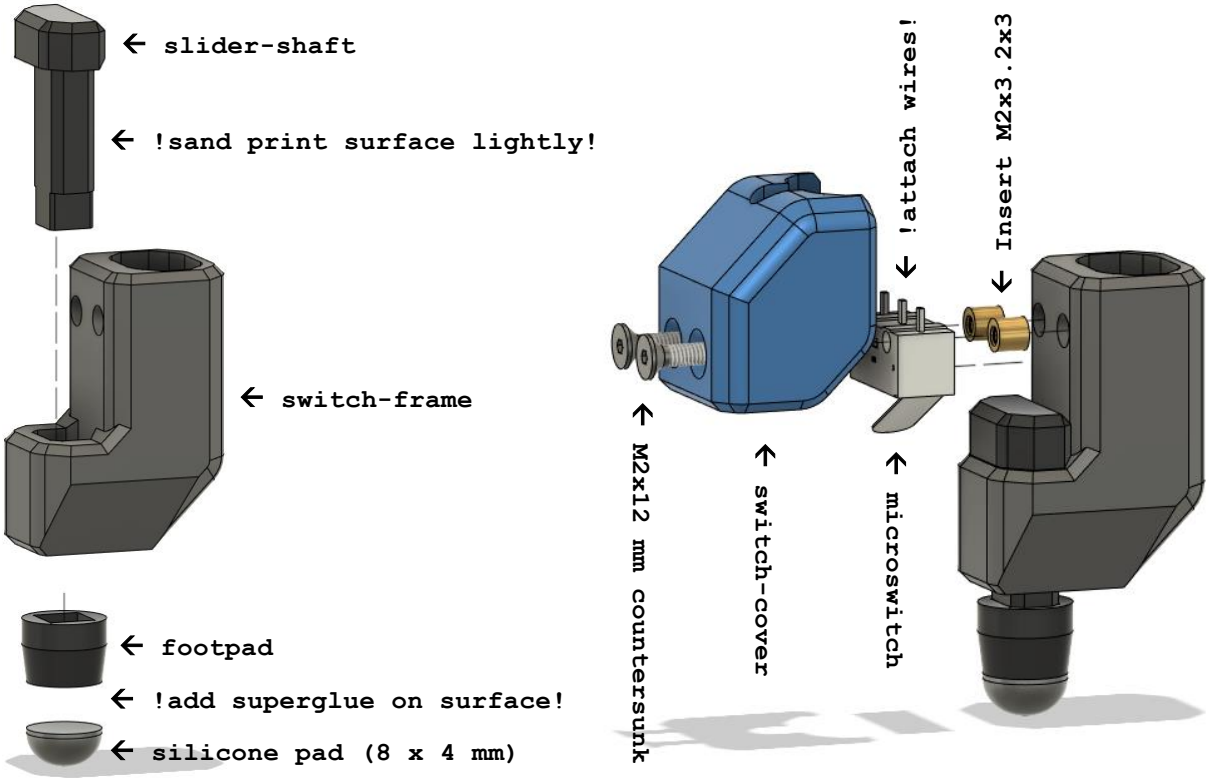


!Depending on your 3d printer rolarances, you may need to glue the tip-shaft to the tip-frame!

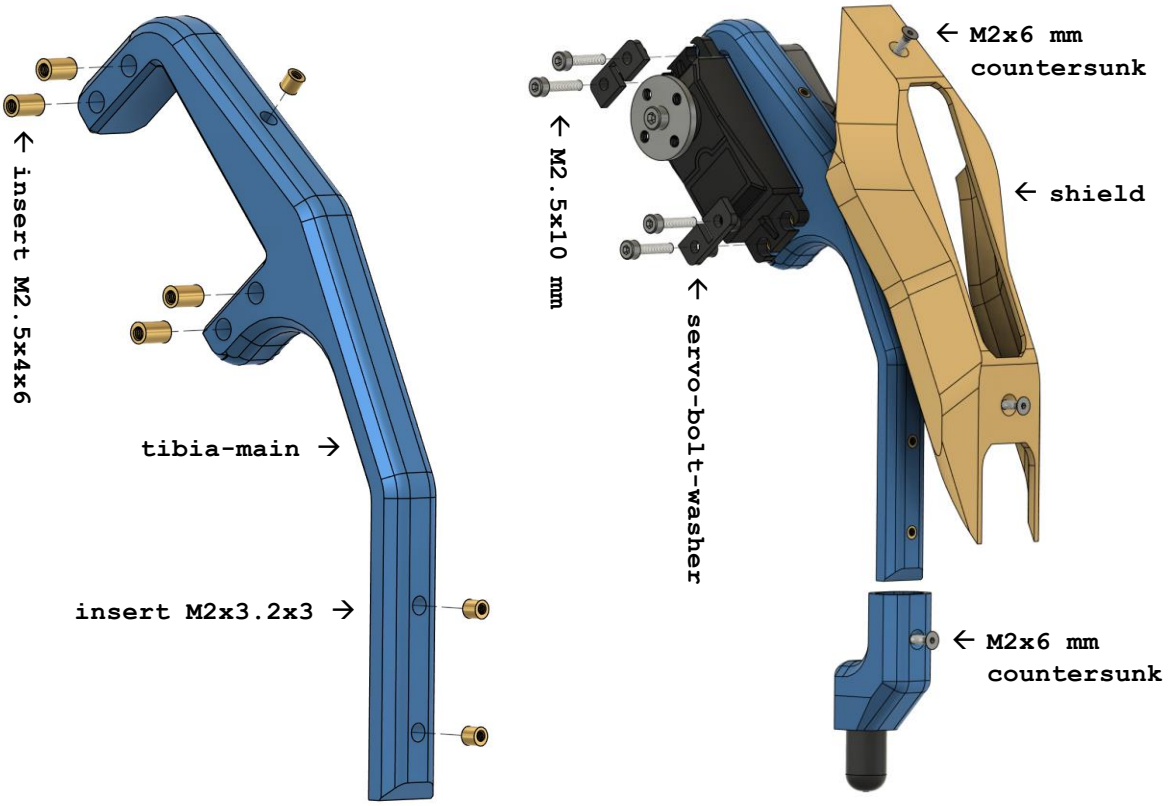
!Note, the rubber vacuum cap can be replaced with a 6mm ID clothes hanger rubber end cap!

!The sensor tip assebly requires a redisign to allow and end cap / vacuum cap to be used for the contact point, the glued silicone pad has poor longevity (they fall off over time).

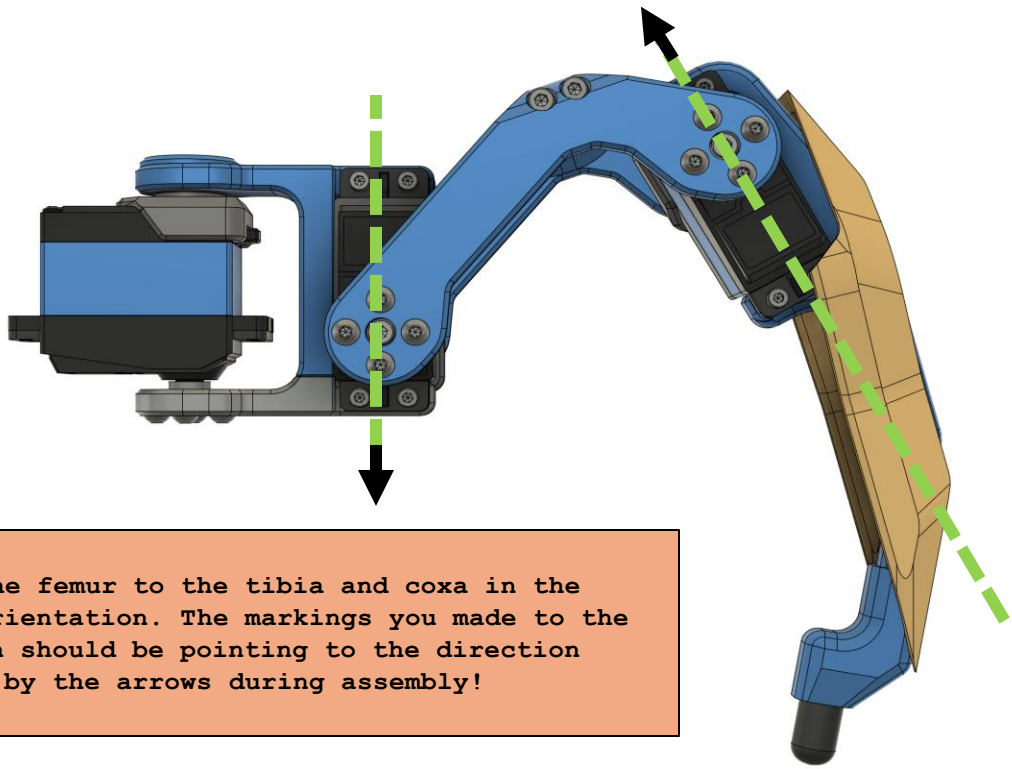
D-2. Sensor tip assembly × 6 (OPTIONAL)



E. Tibia assembly × 6

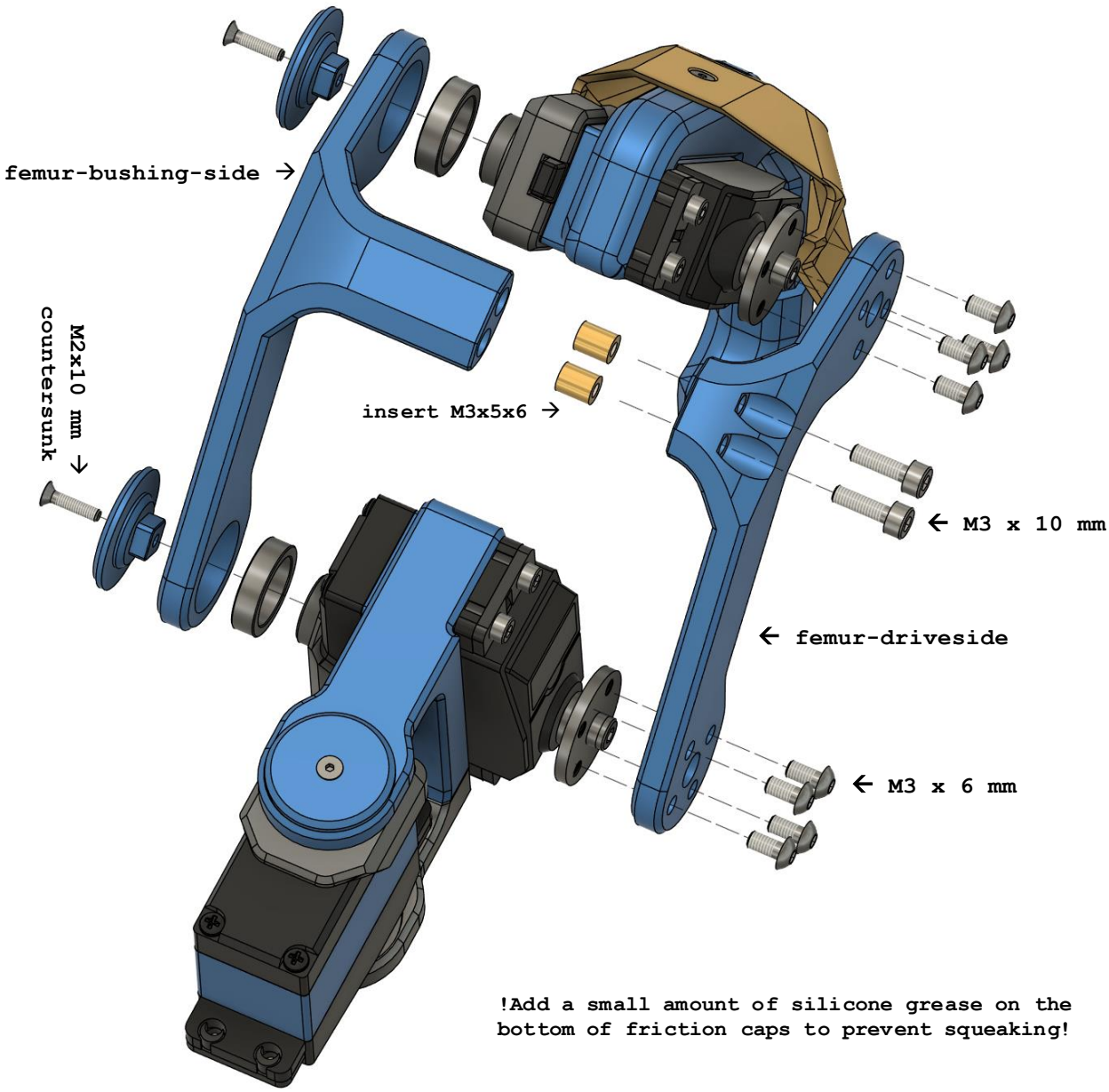


F. Leg alignment



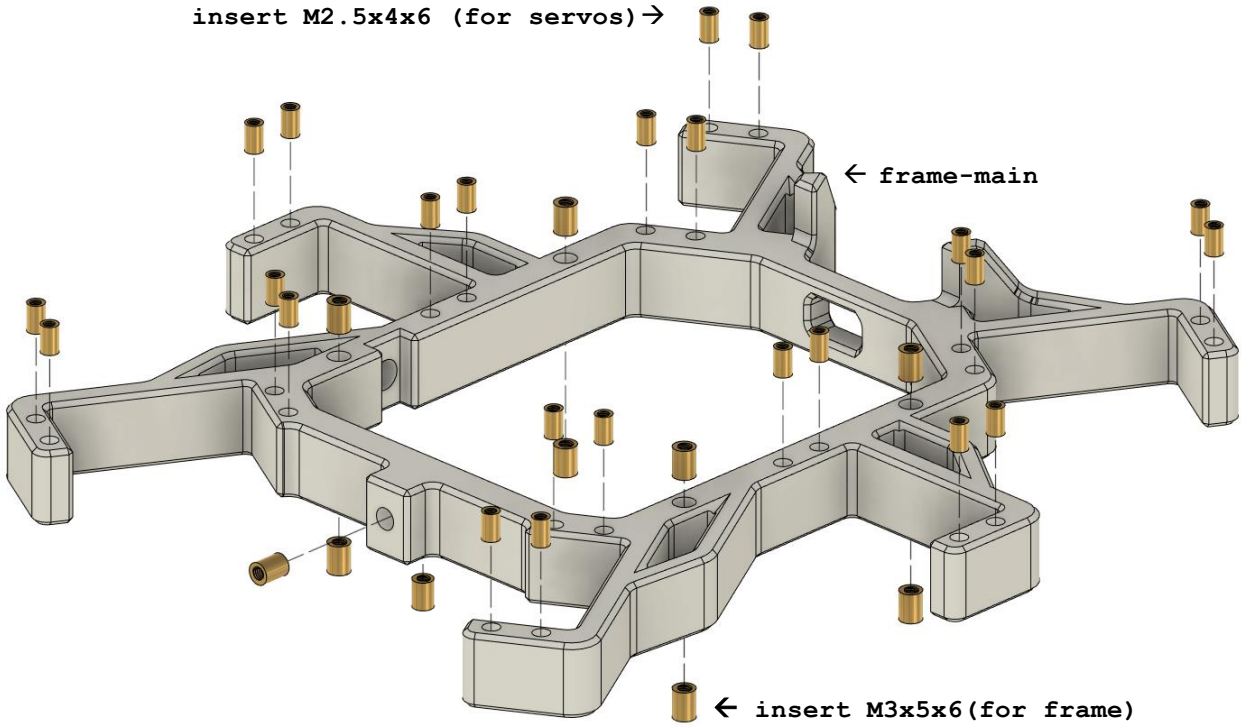
!Attach the femur to the tibia and coxa in the correct orientation. The markings you made to the servo horn should be pointing to the direction indicated by the arrows during assembly!

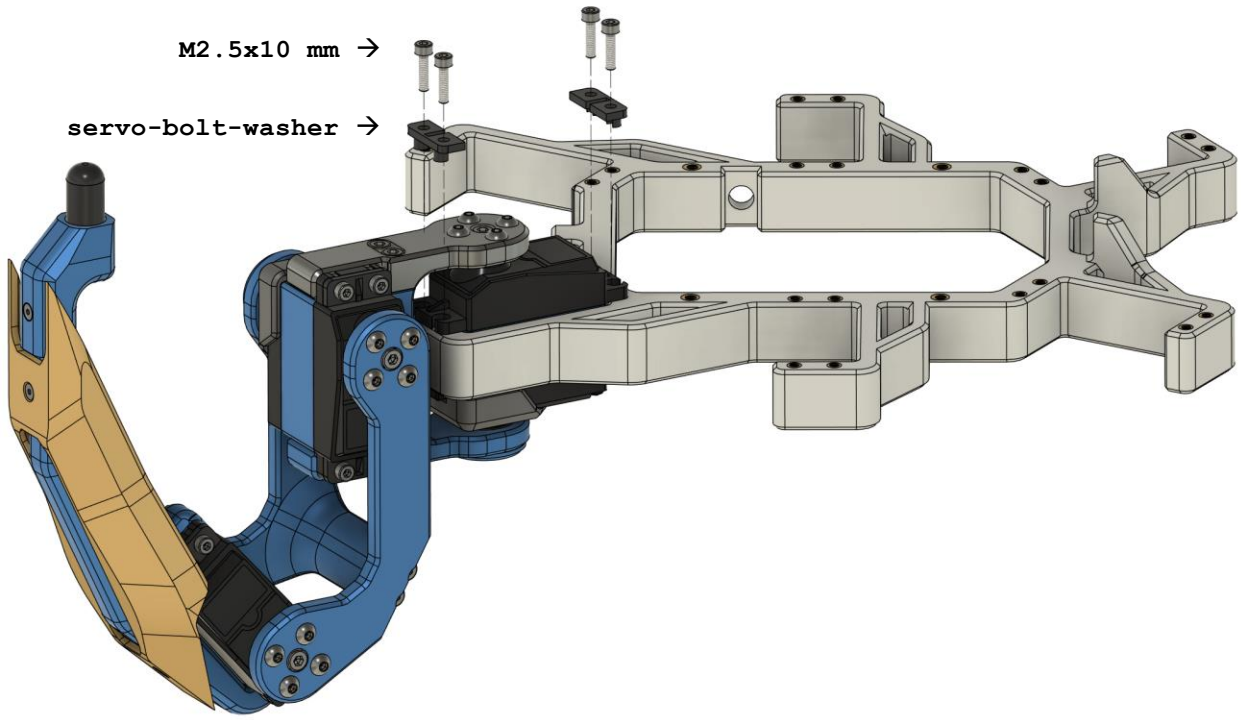
!NOTE: servos need to be calibrated before attaching the coxa, femur or tibia. Number and record the calibration values for your servos (Step A) before proceeding with assembly.



!Add a small amount of silicone grease on the bottom of friction caps to prevent squeaking!

H. Frame assembly



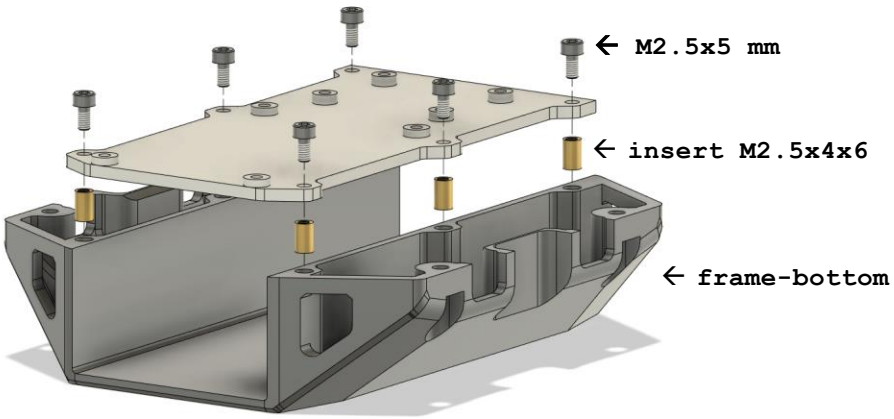


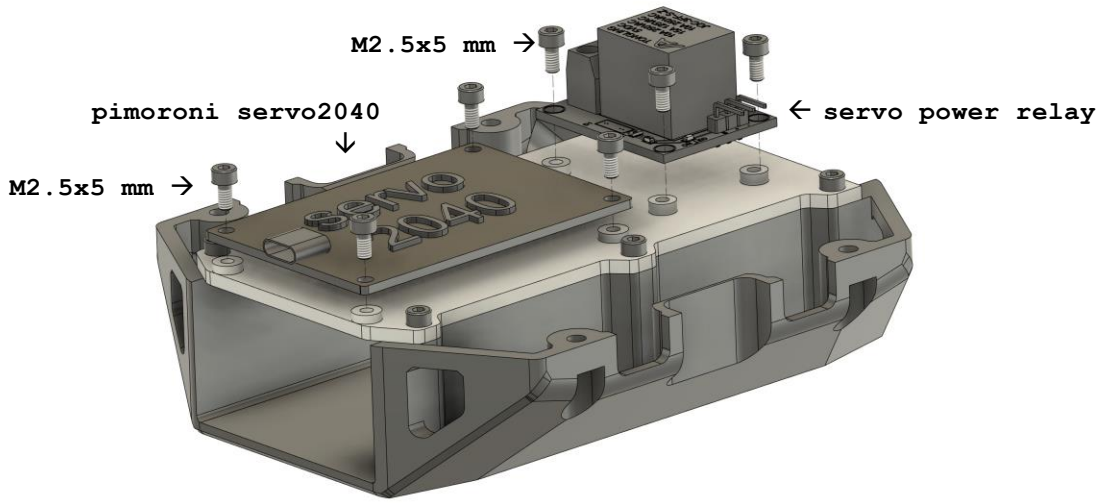
!Attach all six legs before attaching the bottom cover. Attaching servos after is possible, but requires bending the body frame to fit the servos!

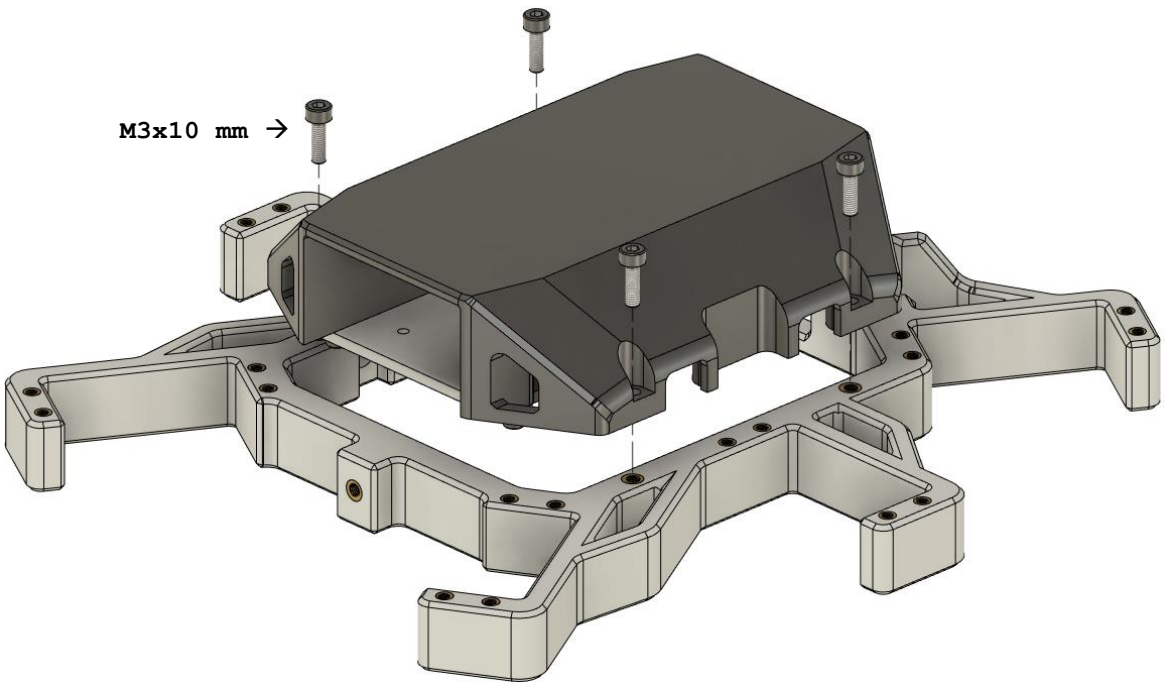
!When attaching legs, the drive side of the servos should face towards the back of the hexapod!

H. Bottom cover & servo controller

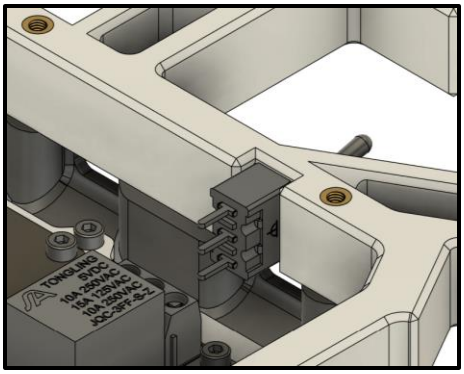
!Attach legs before attaching the bottom cover!







H-1. Servo controller wiring





!CUT THIS!

...or your servo2040 will burn

Rasperry Pi 4/5

usb-c **uArt**

20 AWG

24-26 AWG

16 AWG

Servos:
L-R-C
L-R-F
L-R-T
R-R-C
R-R-F
R-R-T
L-M-C
L-M-F
L-M-T
R-M-C
R-M-F
R-M-T
L-F-C
L-F-F
L-F-T
R-F-C
R-F-F
R-F-T

R1 R2 R3 L1 L2 L3

L/R-R/M/F-C/F/T

L/R
left/right

B/M/F
rear/middle/front

C/F/T
coxa/femur/tibia

!most 5A ubec wiring is too thin out-of-box!

5V 5A UBEC

NC
C
NO

5V relay switch

MTS101 switch

20 AWG

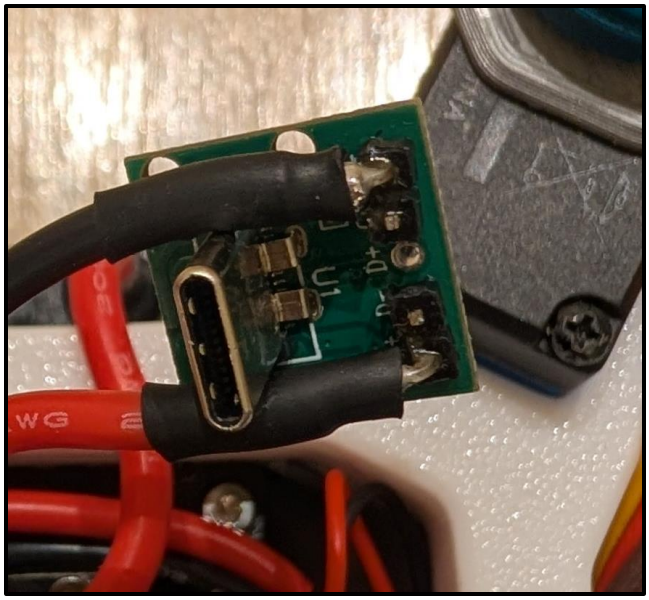
2S (8.4 V) Lipo RC battery

16 AWG

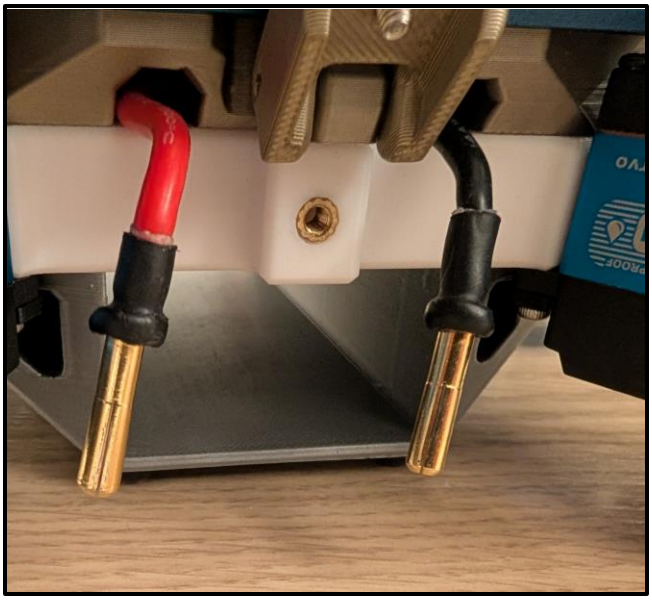
H-2. Servo controller wiring

!Be carefull to never mix/crosswire + and - wires, or stuff will fry!

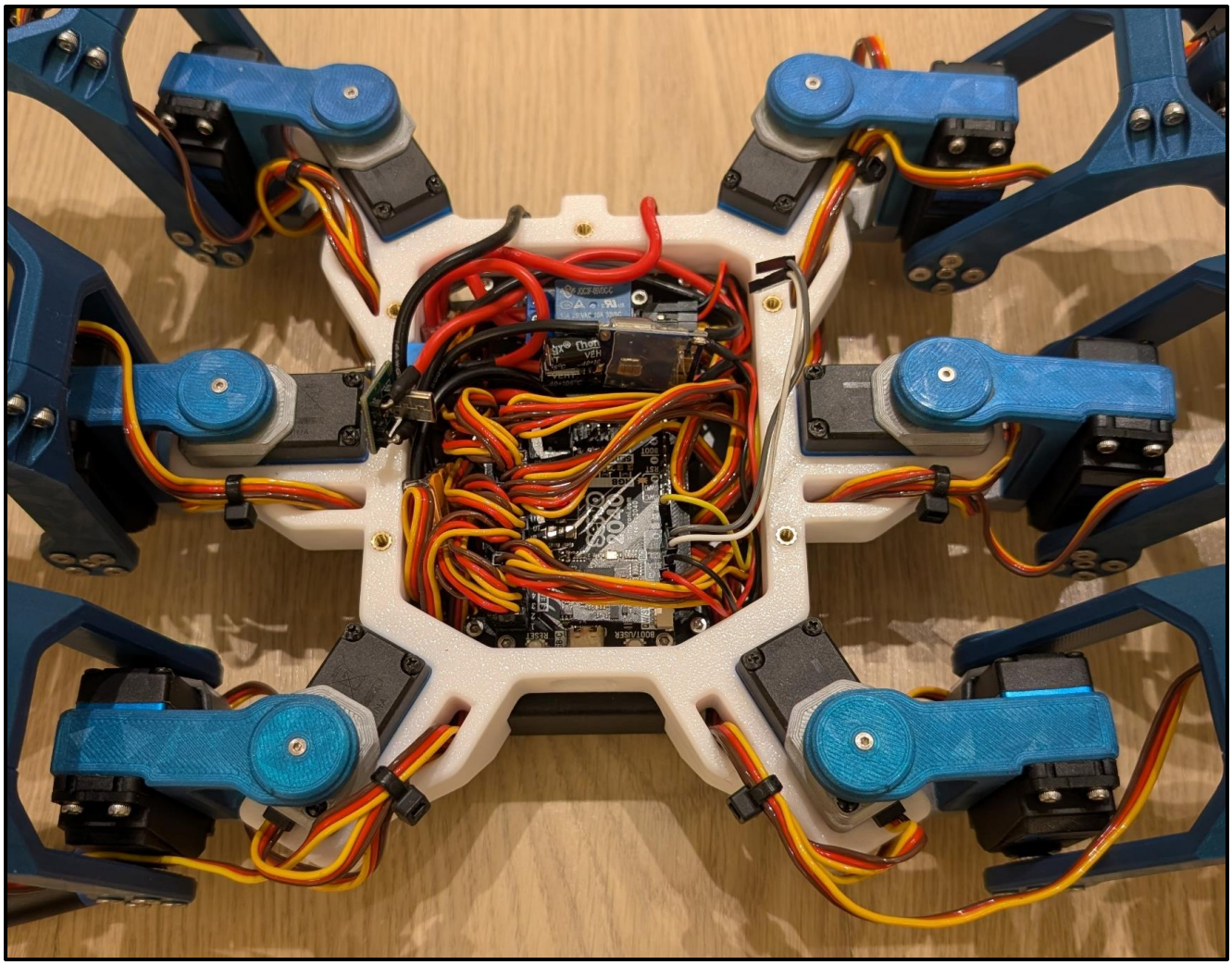
usb-c helper board



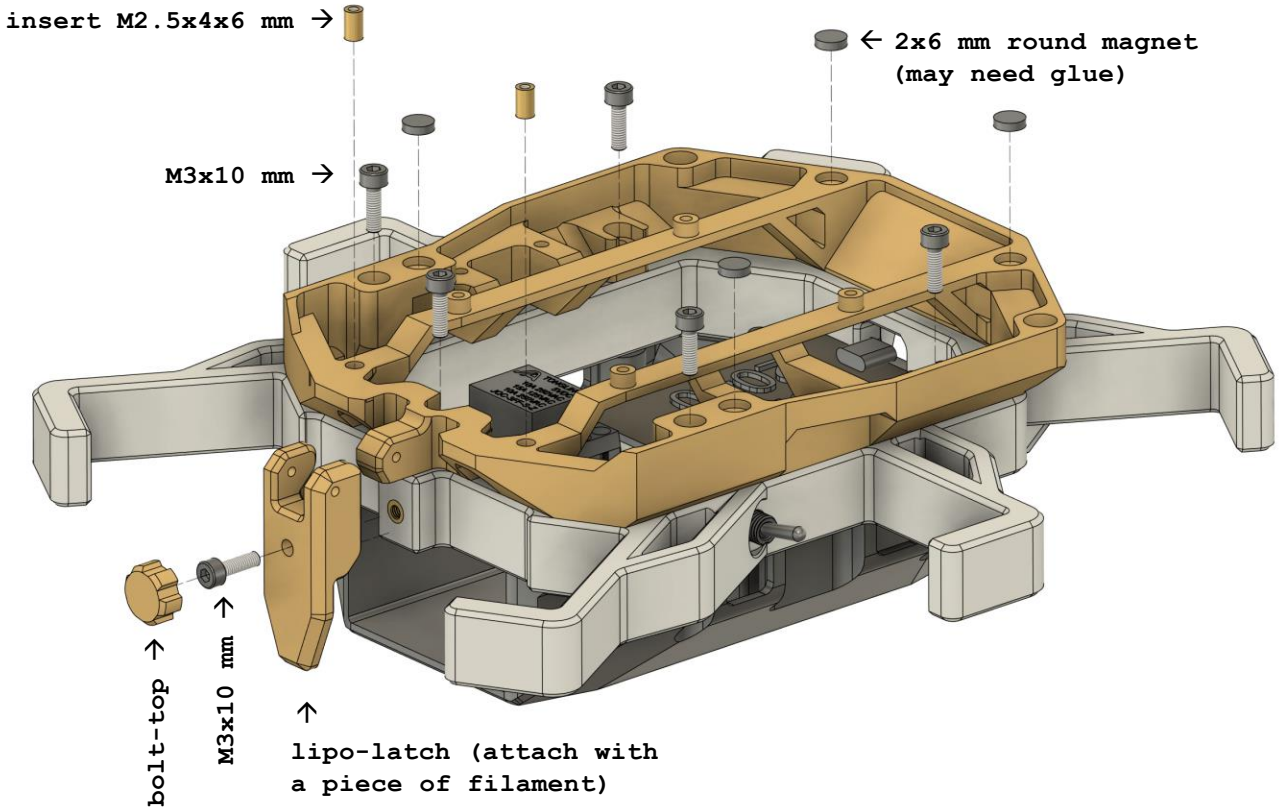
4 mm Lipo connectors

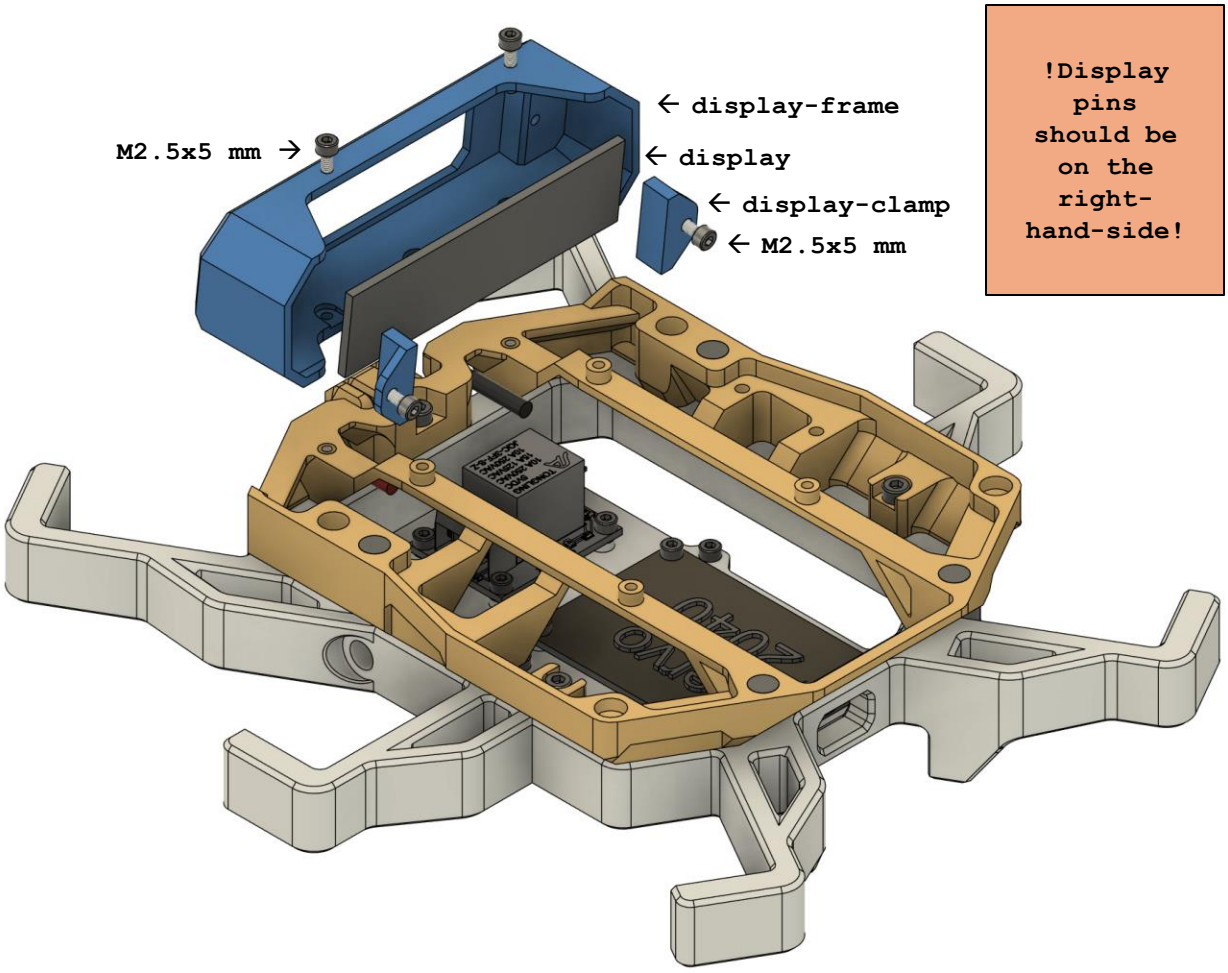


Embrace the spaghetti!



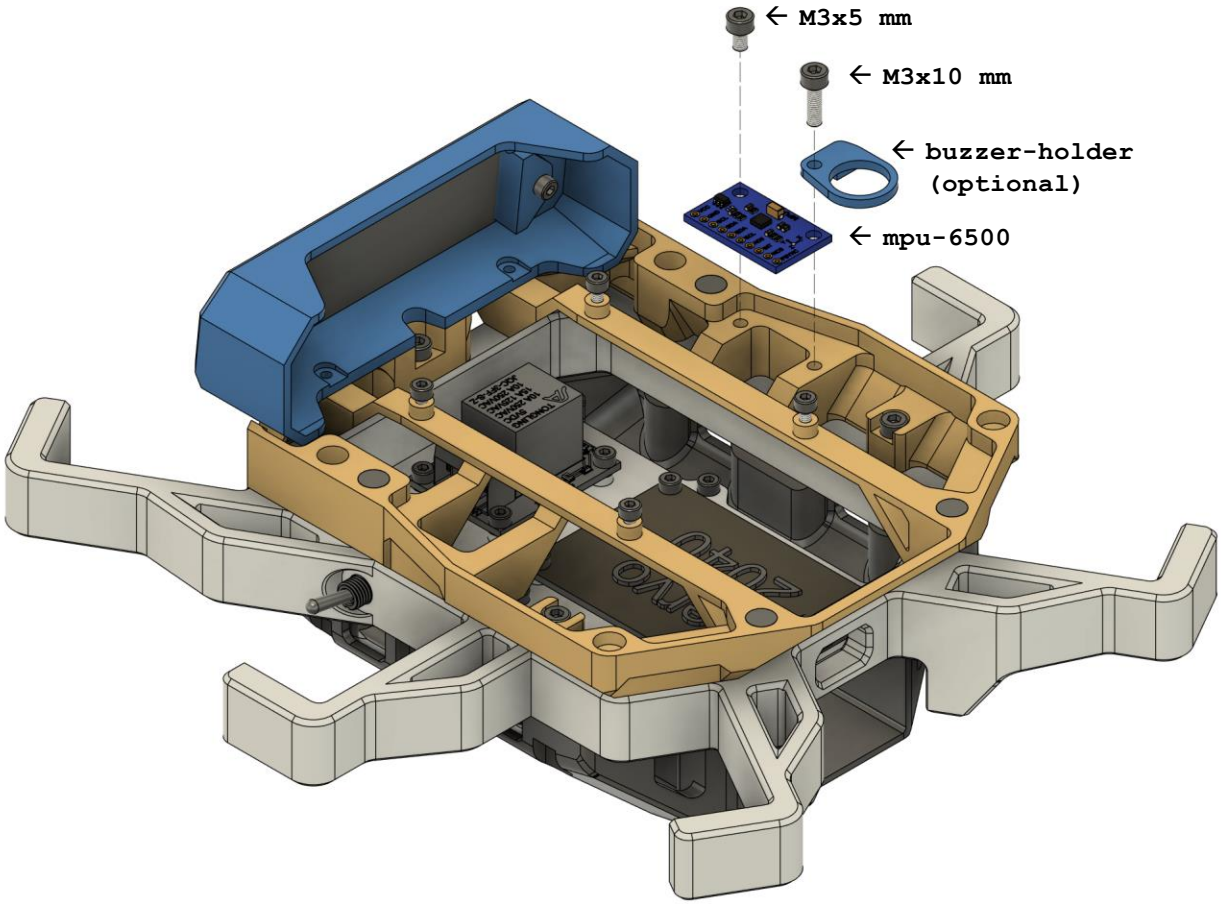
H. Raspberry Pi mounting assembly

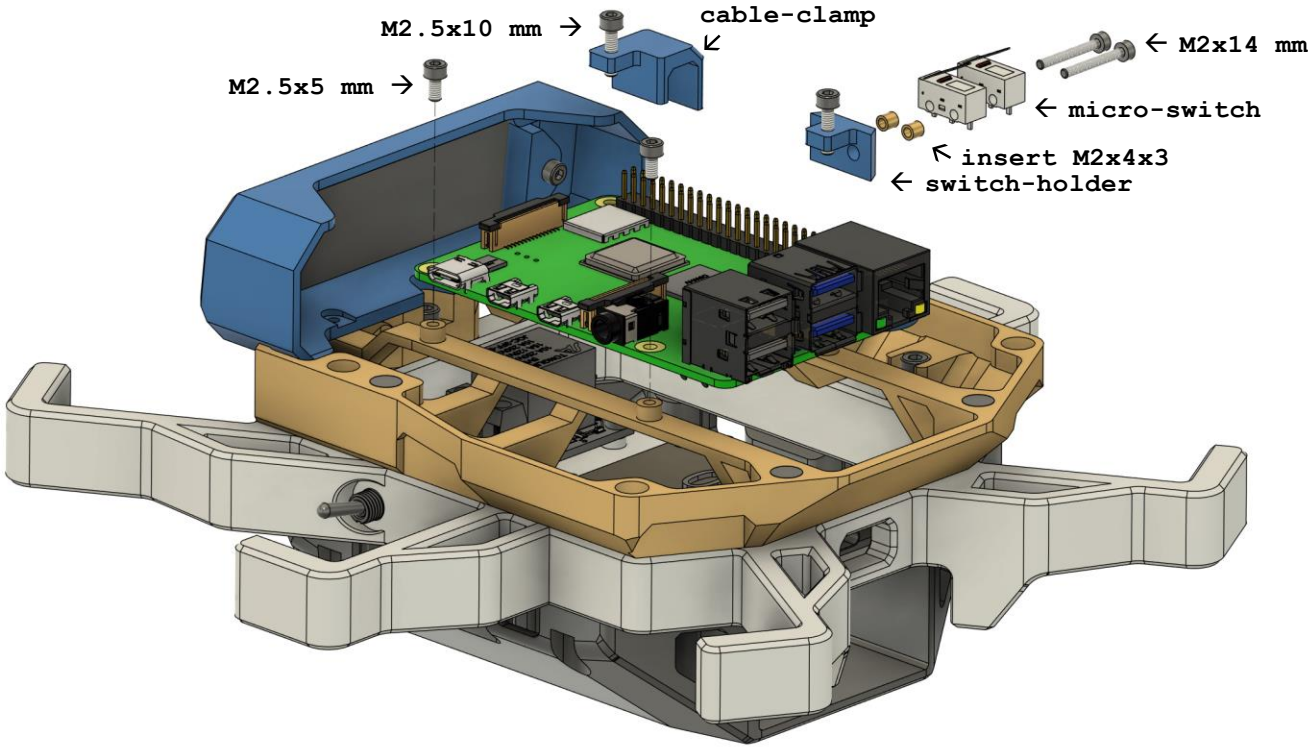




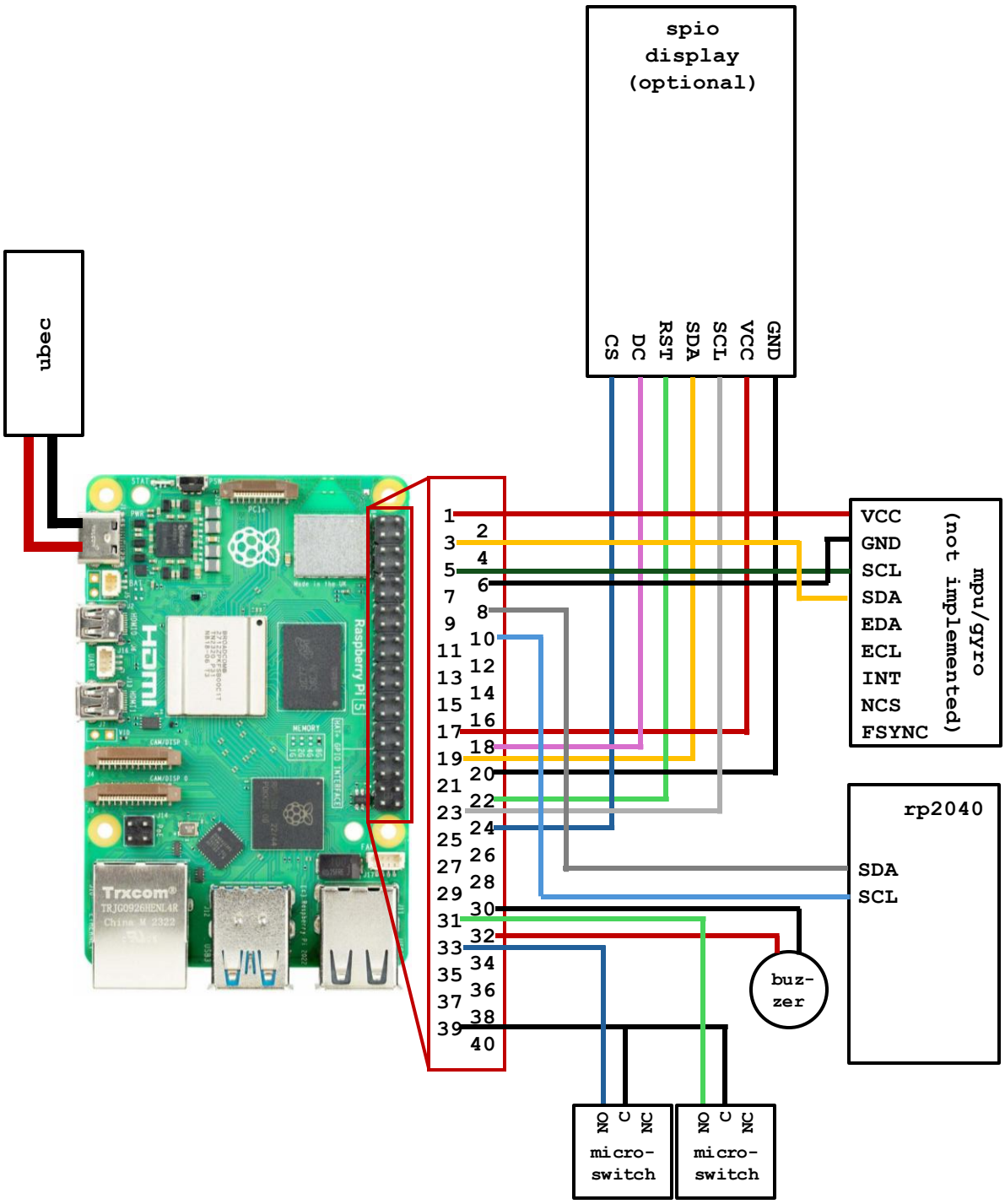
!Display pins should be on the right-hand-side!

H. Raspberry Pi mounting assembly





H-1. Servo controller wiring



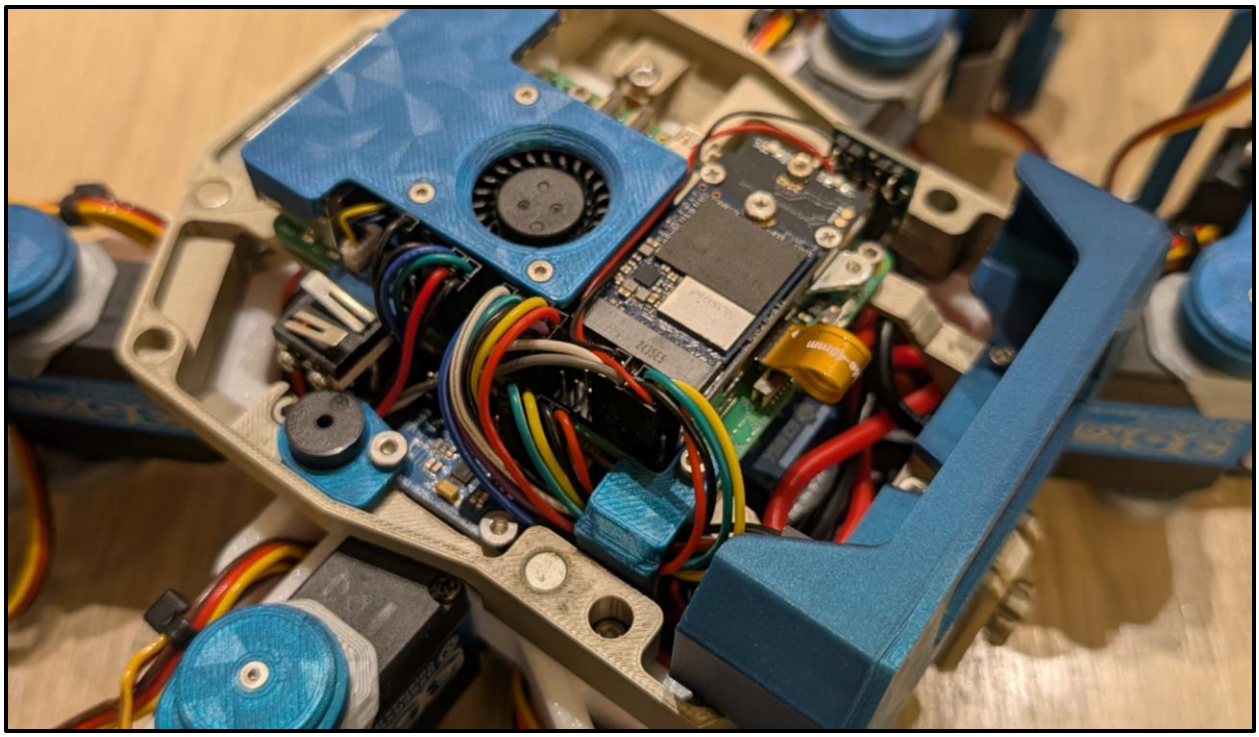
!The hexapod can be ran and operated without the buzzer, micro-switches, the mpu and the display! All of those are technically optional!

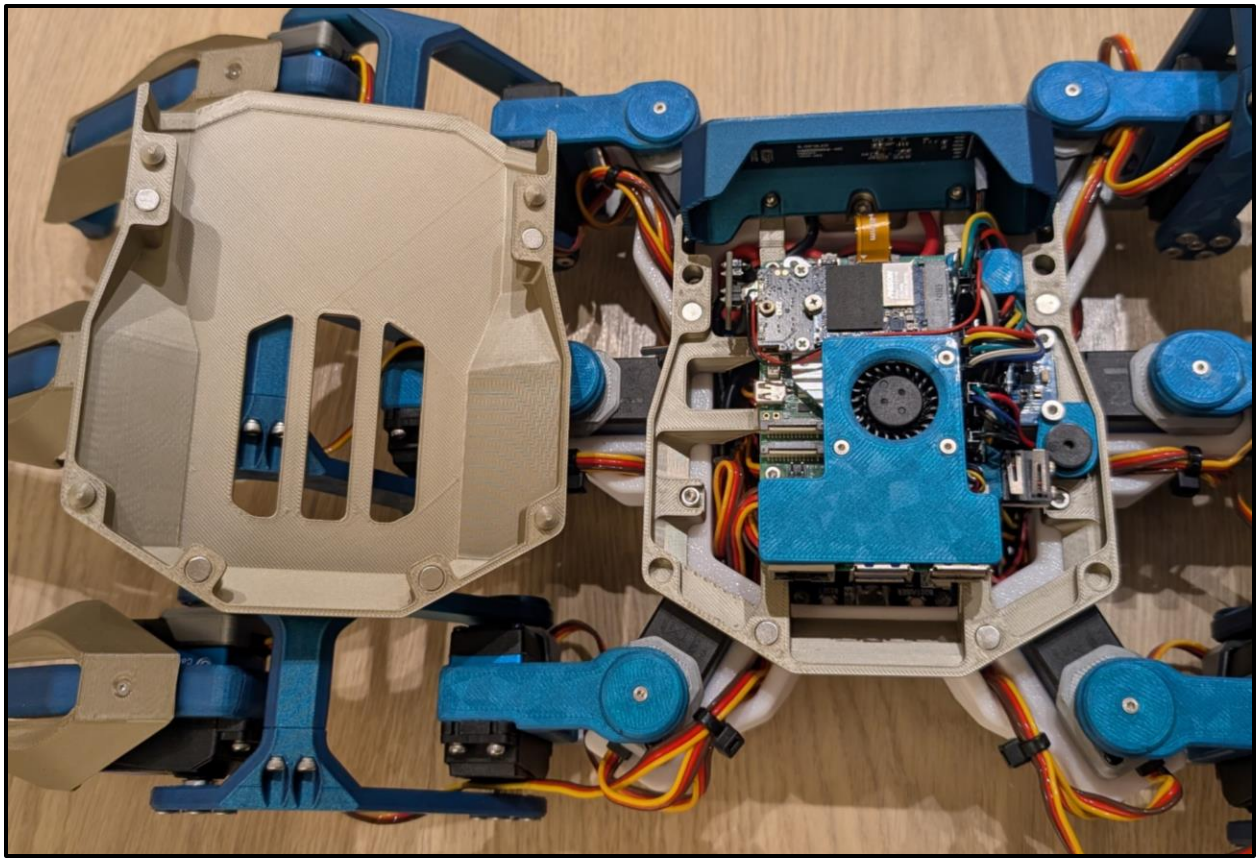
!Display + buttons are used to manage bluetooth connections, wifi setting, etc. These can also be managed via ssh from a terminal!

!20 AWG wires from usb-c to ubec, otherwise regular gpio jumper wires can be used!

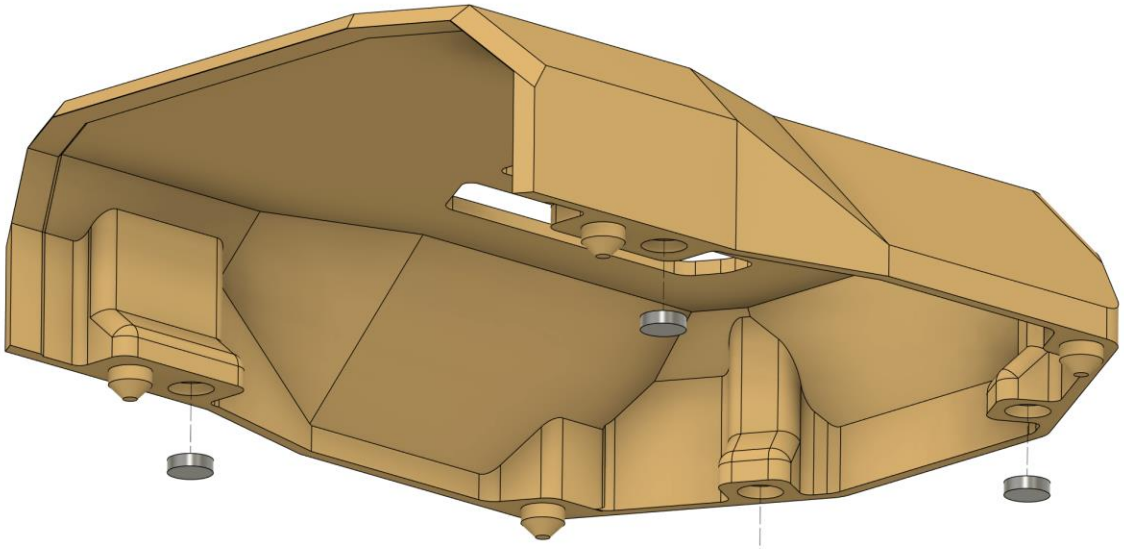
H-1. Servo controller wiring

!As pictured: raspberry pi 5, waveshare PCI-e to M.2 adapter with fan, 32GB 2230 ssd, mpu-6500. NOTE: Wiring color don't match the drawing!





I. Top cover



2x6 mm round magnet(may need glue) →



J. Complete!



!As pictured: blue – Polymaker starlight neptune PLA, gold – Bambu labs PLA metal iridium gold, grey – Bambu labs Grey PLA Basic, white – Sunlu white PLA!